Supplementary Material
bioMoR: Biology-Guided Mixture-of-Recursions for Efficient and Interpretable Genomic Learning

Anonymous Submission

Model	Prostate	BLCA	STAD	PM	PC	3M
Vanilla	64.6±9.0	78.8±2.2	66.7±2.6	57.0±6.3	89.3±2.5	88.7±2.2
Recursive K=2	65.0±8.0	78.0±2.7	66.7±6.4	52.5±8.9	88.6±4.6	90.6±3.9
Recursive K=3	64.6±6.8	78.4±2.1	64.4±5.8	54.3±9.2	90.3±2.1	89.3±2.5
Recursive K=4	60.3±8.2	74.6±5.2	65.5±5.5	51.8±5.9	87.3±3.9	88.2±3.5
MoR-Expert K=2	65.8±8.6	80.3±1.2	66.3±4.0	57.5±3.8	88.1±4.8	88.7±3.6
MoR-Expert K=3	62.3±7.8	78.2±3.6	68.3±3.7	54.3±6.9	89.7±2.0	89.3±5.9
MoR-Expert K=4	63.6±8.9	77.4±3.6	67.0±3.9	52.9±5.2	88.9±3.0	90.6±3.7
Token-choice K=2	65.7±8.2	78.3±3.9	66.6±6.0	58.1±5.2	87.9±2.7	87.9±1.6
Token-choice K=3	71.2±5.2	76.9±3.9	65.3±8.1	52.6±7.9	88.0±4.5	89.3±3.2
Token-choice K=4	70.0±7.6	78.4±3.0	65.8±5.0	55.7±5.8	87.2±3.0	89.6±3.6

Table 1: **PATH-comparable positive-class F1 on the multi-omics cohorts.** Each cell is the threshold-calibrated positive-class (progression: metastatic / late-stage) F1, mean $\pm$ SD over the folds (%), for the full efficiency ladder — the positive-class counterpart of the macro-F1 Table **??**. Following the PATH protocol, the operating threshold is tuned on validation to maximise positive-class F1 and applied to the held-out test fold, with AUROC-based model selection; all other settings match Table **??**. PM/PC are the pan-cancer mutation+CNV and expression cohorts.

This supplement collects the additional tables and derivations referenced in the main paper. Unless stated otherwise, the auxiliary empirical tables here use a 3-seed held-out protocol; the 5-fold cross-validation results reported in the main paper are the authoritative numbers, and we include these analyses for completeness and transparency.

A PATH-Comparable Positive-Class F1

B Model-Size Scaling

C Dataset Details

We use 8 genomap single-cell datasets (Islam and Xing 2023) converted to plain CSV (expression + labels + stratified split) – Baron, Lung, Muraro, Oesophagus, Segerstolpe, Spleen, T-cell and Xin – 3 Reactome/P-NET multi-omics cohorts (prostate, BLCA, STAD) whose fixed Reactome pathway tokens pool mutation, copy-number and expression channels. Per-dataset sample, feature and class counts are read directly from the result files and summarised in Table 8.

	Single-cell			Multi-omics (P-NET)		
d_{model}	Vanilla	MoR-gen.	bioMoR	Vanilla	MoR-gen.	bioMoR
96	63.6±5.9	63.4±4.1	74.8±2.0	52.1±5.1	55.6±9.0	56.2±4.3
136	65.1±5.0	62.4±5.6	75.8±3.8	51.2±7.7	66.0±9.9	53.5±5.6
192	61.8±5.1	62.8±4.3	76.2±3.3	42.5±6.1	*run…*	54.8±4.2
272	60.7±5.8	63.5±4.6	75.4±3.5	52.8±4.8	33.5±6.4	57.0±2.4
352	59.7±5.6	61.4±6.5	77.1±4.0	48.6±8.0	47.3±4.8	56.1±4.5

Table 2: **Model-size scaling.** Macro-F1 (mean±SD) across a width sweep ($d_{\text{model}} \in \{96, \ldots, 352\}$) for Vanilla, MoR-general and bioMoR, averaged over the 8 single-cell and 3 P-NET suites. bioMoR holds a flat, high accuracy across widths while the vanilla stack—at $4\times$ the recursion-stack parameters—degrades.

Dataset	M=128	M=256	M=512	M=1024	M=2048	Linear (all feat.)
Baron	79.1 ± 2.8	78.1 ± 1.4	79.6 ± 4.4	82.6 ± 2.0	80.4 ± 2.1	91.1 ± 2.0
Lung	79.1 ± 1.2	80.2 ± 0.8	79.8 ± 1.9	81.3 ± 2.0	81.1 ± 1.9	74.8 ± 0.2
Muraro	75.3 ± 6.4	72.4 ± 2.7	68.3 ± 3.3	75.7 ± 4.5	73.8 ± 6.0	95.7 ± 2.2
Oeso.	69.6 ± 1.4	69.8 ± 0.6	71.5 ± 1.6	73.6 ± 0.5	72.2 ± 1.6	69.5 ± 1.1
Seger.	71.4 ± 5.3	70.3 ± 4.1	73.8 ± 5.4	69.4 ± 14.7	68.3 ± 3.3	89.5 ± 5.0
Spleen	58.2 ± 0.9	58.6 ± 1.6	59.9 ± 0.8	61.1 ± 2.3	61.6 ± 4.0	64.7 ± 0.1
T-cell	68.9 ± 1.0	69.9 ± 0.3	71.5 ± 0.4	71.8 ± 2.3	73.0 ± 0.3	78.1 ± 0.3
Xin	64.7 ± 9.4	64.8 ± 6.0	63.7 ± 10.3	64.1 ± 0.6	62.1 ± 2.8	97.4 ± 0.6
Mean macro-F1	**70.8**	**70.5**	**71.0**	**72.5**	**71.6**	**82.6**

Table 3: **Marker-budget headroom (3-seed held-out).** Single-cell macro-F1 as the marker budget M grows, against the full-feature linear baseline. Even when every feature is a marker, a residual to the linear model remains, so the gap is architectural rather than a token-budget limitation.

D Effective-FLOPs Accounting

The compute numbers report per-sample FLOPs of the recursive transformer stack, the only component routing changes. One application of the shared block to a tokens costs $\phi(a) = 4a^2d + 4\,a\,d\,d_{\text{ff}}$; the nominal fixed-depth cost is $\Phi_{\text{nom}} = K\,\phi(M)$ and the effective cost sums one block over the tokens active at each step, $\Phi_{\text{eff}} = \sum_{t=1}^{K} \phi(a_t)$, where a_t is the mean number of markers the expert-choice funnel keeps at step t. Gene embedding and marker selection are $\mathcal{O}(Nd)$, identical across routing modes, and excluded from this stack-level comparison.

End-to-end accounting. For completeness we account for the whole forward pass, not only the stack. Three parts scale with the full gene count N: gene embedding $\Theta(Nd)$; the

Dataset	Learned (random init)	+ bio warm-start	+ stronger init	+ annealed anchor
Baron	82.3 ± 2.2	80.9 ± 3.8	81.6 ± 1.8	66.8 ± 6.2
Lung	79.5 ± 1.4	79.8 ± 0.8	78.4 ± 0.6	70.0 ± 2.7
Muraro	74.5 ± 5.4	75.3 ± 4.9	76.6 ± 5.4	75.9 ± 9.3
Oeso.	69.3 ± 0.7	71.1 ± 0.9	70.2 ± 1.0	57.4 ± 2.6
Seger.	74.0 ± 5.3	69.9 ± 7.4	74.5 ± 5.7	53.0 ± 5.0
Spleen	59.0 ± 0.2	59.1 ± 0.7	58.7 ± 1.2	52.5 ± 1.2
T-cell	68.9 ± 0.7	69.1 ± 1.1	69.0 ± 0.6	54.7 ± 2.4
Xin	69.2 ± 5.0	67.2 ± 8.4	68.5 ± 6.1	69.5 ± 7.7
Mean macro-F1	**72.1**	**71.5**	**72.2**	**62.5**
Δ vs. random	+0.0	−0.6	+0.1	−9.6

Table 4: **Stress-testing the biological warm-start** (single-cell macro-F1, mean±std). From a randomly-initialised learned graph we add a biological warm-start, then a stronger init, then an annealed anchor $\lambda(t)\|A_{\text{learned}} - A_{\text{bio}}\|_F^2$ that keeps the graph near biology early. The Δ row is the mean gain over random init; forcing biology in more strongly does not beat learning the graph from data.

Cohort	bioMoR (mut+CNV)	Geneformer	scGPT
Prostate	78.2 ± 2.2	78.2 ± 3.3	40.1 ± 0.0
BLCA	40.9 ± 1.9	47.7 ± 3.7	40.4 ± 0.0
STAD	52.2 ± 6.8	46.1 ± 5.5	35.7 ± 0.0
Mean	**57.1**	**57.3**	**38.7**

Table 5: **bioMoR vs. gene-vocabulary foundation models** on the P-NET cohorts (macro-F1, mean±std over seeds). Geneformer (fine-tuned) and scGPT (frozen embedding + logistic probe) are mapped onto each cohort's HUGO gene symbols with a per-gene mutation/copy-number alteration burden, on the same splits as bioMoR. Bulk DNA-alteration input is out-of-distribution for these single-cell-RNA models.

cross-attention marker router, whose M queries attend over all N gene keys at $\Theta(MNd)$; and the optional gene-graph smoother, which at rank r costs $\Theta(Nr)$ (never the N^2 dense form). Everything after marker selection – the recursive stack, pooling and head – scales with the compressed budget $M \ll N$ at $\Theta(M^2 d)$ per pass. So the router's $\Theta(MNd)$ term dominates the N-scaling and is *linear* in N (versus the $\Theta(N^2 d)$ self-attention a full-gene transformer would pay), while the quadratic cost is paid only on the M markers. The architecture ablations of the main-paper efficiency ladder vary only the stack, so the stack-level ratios there are the correct comparison for the routing/sharing claims; the router and embedding terms are shared by every variant. Measured wall-clock and peak-memory profiling on matched hardware is a straightforward addition we leave to the camera-ready.

E Theoretical Foundation of the Router

This appendix gives the mathematical and biological grounding for bioMoR's biology-informed router.

Routing as conditional computation. The router implements *conditional computation*: a learned policy routes each token to a token-specific amount of compute, the gating principle of sparsely-gated mixtures of experts (Shazeer et al. 2017), reused by Mixture-of-Depths (Raposo et al. 2024) and Mixture-of-Recursions (Bae et al. 2025); the "experts"

Depth K	Shared (ours)	Independent	Reduction
1	74,880	74,880	1×
2	74,880	149,760	2×
3	74,880	224,640	3×
4	74,880	299,520	4×
6	74,880	449,280	6×
8	74,880	599,040	8×

Table 6: **Parameter reduction.** One shared block versus K independent layers at matched width: an exact $K\times$ reduction, present before any training.

Dataset	M=16	M=32	M=64	M=128	M=256
Single-cell (genomap)					
Baron	33.7 ± 7.1	38.5 ± 2.1	53.9 ± 1.9	62.6 ± 5.1	66.9 ± 3.4
Lung	30.8 ± 5.9	38.1 ± 1.7	61.5 ± 1.9	71.4 ± 1.1	78.4 ± 0.8
Muraro	55.8 ± 3.6	59.4 ± 1.3	67.9 ± 5.6	68.5 ± 4.1	76.2 ± 5.2
Oeso.	23.1 ± 1.3	30.2 ± 3.9	42.0 ± 2.4	53.9 ± 4.4	63.2 ± 2.4
Seger.	41.2 ± 12.1	45.3 ± 5.6	59.3 ± 4.0	59.1 ± 1.6	55.4 ± 7.0
Spleen	18.7 ± 2.4	26.8 ± 3.0	38.1 ± 2.3	48.1 ± 5.0	59.4 ± 0.9
T-cell	25.5 ± 3.2	36.6 ± 5.4	43.1 ± 5.2	47.4 ± 2.2	61.5 ± 1.7
Xin	50.5 ± 5.9	46.2 ± 1.6	60.8 ± 5.6	64.1 ± 3.3	72.4 ± 11.7
Multi-omics (Reactome/P-NET)					
Prostate	74.3 ± 3.0	74.7 ± 2.7	78.0 ± 3.0	76.7 ± 2.6	77.3 ± 0.2
BLCA	49.8 ± 2.2	46.0 ± 3.1	49.3 ± 2.5	48.1 ± 6.8	55.7 ± 3.5
STAD	54.2 ± 3.1	51.0 ± 2.2	44.0 ± 5.6	51.2 ± 2.8	49.4 ± 4.1
Mean macro-F1	**41.6**	**44.8**	**54.4**	**59.2**	**65.1**

Table 7: **Marker-token budget.** Macro-F1 (mean±std over 3 seeds) as the marker budget M shrinks from 256 to 16. A few dozen to a few hundred tokens recover most of the full-gene accuracy. Bottom row: mean over all 11 datasets.

here are recursion *depths* of one shared block, which couples adaptive computation (Graves 2016) to weight sharing.

The differentiable handle. The discrete top-k has zero gradient almost everywhere, so bioMoR keeps the soft probability of the chosen route as a multiplicative *gate* on the block output, $\mathbf{o}_m = g_m\, f_\theta(\mathbf{h}_m)$ with $g_m = \sigma(\tilde{r}_m)$. Because g_m is smooth in $\mathbf{w}_r$, the chain $\mathcal{L} \leftarrow \mathbf{o}_m \leftarrow g_m \leftarrow \mathbf{w}_r$ is unbroken: the hard choice routes, the soft gate carries the gradient. The biological prior of the routing logit of the main paper (Eq. 1) is an additive constant in this logit, so it shifts the decision without breaking this path.

Biological foundation of the prior. Two facts from single-cell biology motivate the prior. A small set of *marker* genes carries most discriminative signal (Ianevski, Giri, and Aittokallio 2022; Franzén, Gan, and Björkegren 2019; Hu et al. 2023), so compute should be allocated unevenly. And gene co-expression networks are approximately scale-free: a few high-degree *hub* genes (master regulators) exert outsized influence. We operationalise "hub" as eigenvector centrality on the genomap co-expression graph $\mathbf{W}$: the leading eigenvector $\mathbf{v}$ of $\mathbf{W}\mathbf{v} = \lambda\mathbf{v}$ scores each gene by the centrality of its neighbours, recursively, and we z-score it to form π.

Empirical-Bayes reading. The routing logit of the main paper (Eq. 1) is a log-linear prior on the routing decision, with

Dataset	Modality	Samples	Features	Classes
Baron	single-cell	8,569	2,000	13
Lung	single-cell	57,020	1,089	25
Muraro	single-cell	2,122	2,000	9
Oeso.	single-cell	87,947	1,089	18
Seger.	single-cell	2,127	2,000	11
Spleen	single-cell	94,129	1,089	28
T-cell	single-cell	24,007	1,089	7
Xin	single-cell	1,492	2,000	4
Prostate	multi-omics	1,011	1,223	2
BLCA	multi-omics	404	1,258	2
STAD	multi-omics	414	1,258	2
PM (`pan_meta`, mut+CNV)	multi-omics	8,893	10,333	2
PC (`pan_meta`, expr)	multi-omics	8,586	10,013	2

Table 8: The 13 datasets used throughout: 8 genomap single-cell suites 3 Reactome/P-NET multi-omics cohorts. Counts are read directly from the result files.

$\beta_t\,\pi_m$ a Gaussian-like prior mean and $\beta_t = \beta_0(1 - \text{progress})$ a shrinkage strength that decays as data evidence accumulates: prior-dominated when the likelihood is uninformative (early training), data-dominated later.

Leakage safety. $\mathbf{W}$ is computed from training-split *expression only*; no cell-type label enters it. The prior therefore injects network topology, not the answer, which is why a gene-discovery claim under this prior is not circular, unlike a prior built from curated cell-type marker lists.

Optional pathway-graph smoothing. The additive bias treats genes independently. Co-pathway genes should route coherently, which one obtains by smoothing the logits over $\mathbf{W}$ with the normalised Laplacian $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$, $\hat{\mathbf{r}}^{(t)} = \tilde{\mathbf{r}}^{(t)} - \gamma\,\mathbf{L}\tilde{\mathbf{r}}^{(t)}$ (graph-Laplacian regularisation): a gene borrows routing strength from its network neighbours. This is one sparse matrix-vector product per step and adds no transformer parameters; we expose it as an option and leave its evaluation to future work.

References

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